

DATASCI 530 – Assignment 4

For this assignment, you will practice the use of regular expressions to extract information from text data. You will be working with real text from bills actions from the 116th Congress of the United States:

https://en.wikipedia.org/wiki/116th_United_States_Congress

For this homework, you will be working with the “bills_actions.csv” dataset in Lecture 10:

https://github.com/alejandrosanchezbecerra/gtm531spring2024/blob/main/Lecture%2010/data_raw/bills_actions.csv

You will be asked to process unstructured text data and extract specific elements of text.

- Report some basic descriptive statistics from the dataset. How many distinct “bill” identifiers are there? What proportion of rows belongs to each category? What proportion belongs to different “main_action”.
- How many House floor actions are associated with bills in the “Senate bills” category?
 - o What proportion of these actions were suspended? What proportion of these actions were suspended as amended?
 - o The House floor actions mention specific people by (Mr., Ms., Mrs., etc). Use regular expressions to extract the names of specific people. Produce a new dataset counting how many bills mention the same person and then extract the unique list of names of the people mentioned. What is the distribution of total bills proposed per person?
- Now focus on a subset of bills whose actions are marked as “Senate amendment proposed (on the floor)”. An important feature of the text is that it can mention one or more senators.
 - o Produce a new dataset counting how many senators are mentioned per bill. This dataset should also include two variables, with the name of the first senator (Senator 1) mentioned in the bill and the name of the second senator (Senator 2). What is the distribution of the number of senators per bill?
 - o With how many different people does each Senator coauthor bills, i.e. how many different people appear as Senator 2 for each Senator 1?
- Briefly describe your insights.

New insight question:

- (f) Compute an additional table/figure not included in the questions above.

(g) What do we learn from it?

Think carefully about how to present your results. To facilitate the process of using regular expressions, you may want to use the following additional information on regular expressions:

<https://www.dataquest.io/blog/regex-cheatsheet/>

<https://www.geeksforgeeks.org/python-regex-cheat-sheet/>

Deliverable:

Submit an HTML file with your results on Canvas. Include Markdown text chunks to write comments on your findings and explain your methodology. You are expected to have a title for your project and a short section at the beginning summarizing the findings. We recommend that you do not print raw data and only print final tables and/or figures.

When submitting your work, imagine that you are sharing your report with colleagues who have asked you to explain and justify the key decisions and findings from your analysis. The file should be professionally presented, self-explanatory, and have clearly labeled sections.

Rubric:

You will be graded both on the code and the writing using the following point system:

Code - Code runs without issues. - Code is appropriate for answering the questions. - File is appropriately organized, with different sections and comments clarifying the definition of variables and dataset names.	2 points
Description of methodology - Key coding decisions are explained via Markdown chunks. - Choices about how to clean the data are explained. - Quality checks are conducted, such as verifying the sample size of merged datasets, checking for the presence/absence of missing values for key variables, and/or computing any relevant descriptive statistics.	2 points
Results and Visualization - Includes a short summary of the findings. - Results are correctly interpreted and easy to follow. - Includes either tables or graphs to display results from different groups.	4 points

- All graphs look professional (have axis and title labels, a suitable color scheme, appropriate font sizes, a legend when needed, etc.). - Tables are properly formatted (e.g., have appropriately rounded values, relabeled columns, captions, and are not too long).	
New Insight - Analysis is distinct from the questions already provided. - Clear interpretation of the new analysis.	2 points